Rate floor stable across a 20× Δt sweep

exp044 · 2 June 2026

The trained networks this entry uses are produced once in the shared training hub, exp022 (Training), and reused here rather than retrained.

Abstract

The Δt audit asks whether the exp025 headline E rate is a physical (Hz) property of the trained network or an artefact of the integration timestep. The rate stays within a 9–14 Hz band across a 20× Δt sweep, accuracy holds at 90.4–91.4%, and the gamma cycle period in physical ms is invariant. The rate is not fully Δt-independent, though: it rises monotonically with the timestep, from ≈ 9.2 Hz at Δt = 0.05 ms to ≈ 13.4 Hz at Δt = 1 ms, so a coarser step inflates the rate while leaving accuracy and cycle period intact.

Method

Checkpoint-backed rates and rasters use the final-epoch checkpoint, matching the epoch-50 endpoint shown in the training histories.

Parameter	Value
Integration timestep Δt	0.05–1.0 ms (swept)
Trial duration T	200 ms
MNIST training pool (7,000 official-training samples)	6,300 optimizer-training / 700 validation; downstream inference uses a fixed 1,000-image subset of the official test partition
Epochs	50

The exp022 hub trains one PING per $\Delta t \in \{0.05, 0.1, 0.25, 0.5, 1.0\}$ ms × seed $\in \{42, 43, 44\}$ = 15 cells; this entry loads them and evaluates. Total physical time $T = 200$ ms is held constant (step count varies 4000 → 200). Batch size `(`

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```
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(
```

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  ),
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    grad_clip: 1.0,
    hidden_sizes: (1024,),
    input_rate: 25.0,
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    lr: 0.0004,
    max_samples: 7000,
    model: "ping",
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    n_in: 784,
    n_inh: 256,
    n_out: 10,
    readout_mode: "mem-mean",
    readout_w_init_mean: 1.12060546875,
    readout_w_init_std: 0.8349609375,
    surrogate_slope: 1.0,
    t_ms: 200.0,
    tau_ampa_ms: 2.0,
    tau_gaba_ms: 6.0,
    trainable_w_ei: false,
    trainable_w_ie: false,
    v_grad_dampen: 1000.0,
```

```
    w_in: (0.9, 0.09000000000000001),
    w_in_initial_zero_fraction: 0.95,
    weight_decay: 0.0,
  ),
  registered_differences: (
    dt_ms: (0.05, 0.1, 0.25, 0.5, 1.0),
    seeds: (42, 43, 44),
  ),
```
`)` is read from and verified across the consumed checkpoint configs. All other PING training parameters are likewise checked for equality; only integration timestep and seed may differ between cells.

Run inference on that held-out subset; report mean E rate (Hz), accuracy, and a single-trial raster from seed 42 per Δt for visual cycle-period inspection.

Results

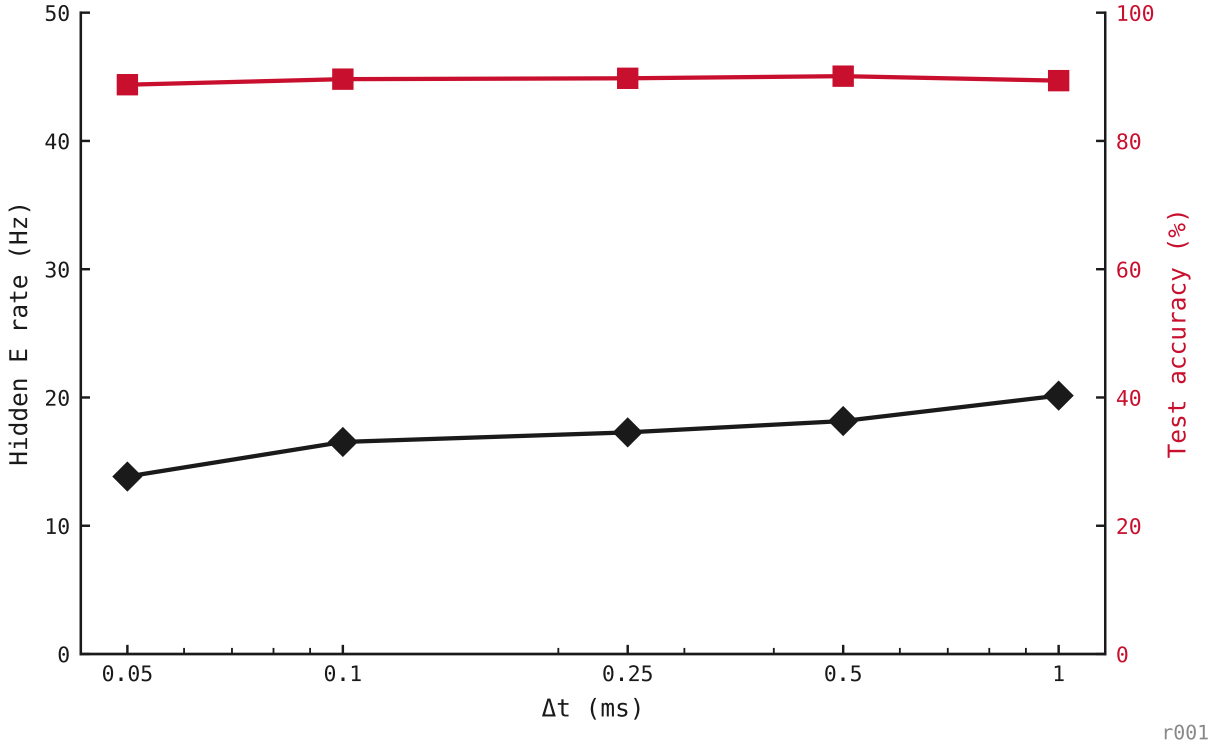


Figure 1: Hidden E rate (black) and test accuracy (red) as Δt varies 20×; markers are per-Δt means over three seeds. Accuracy holds near 91% while the E rate rises monotonically from ≈ 9.2 Hz at Δt = 0.05 ms to ≈ 13.4 Hz at Δt = 1 ms.

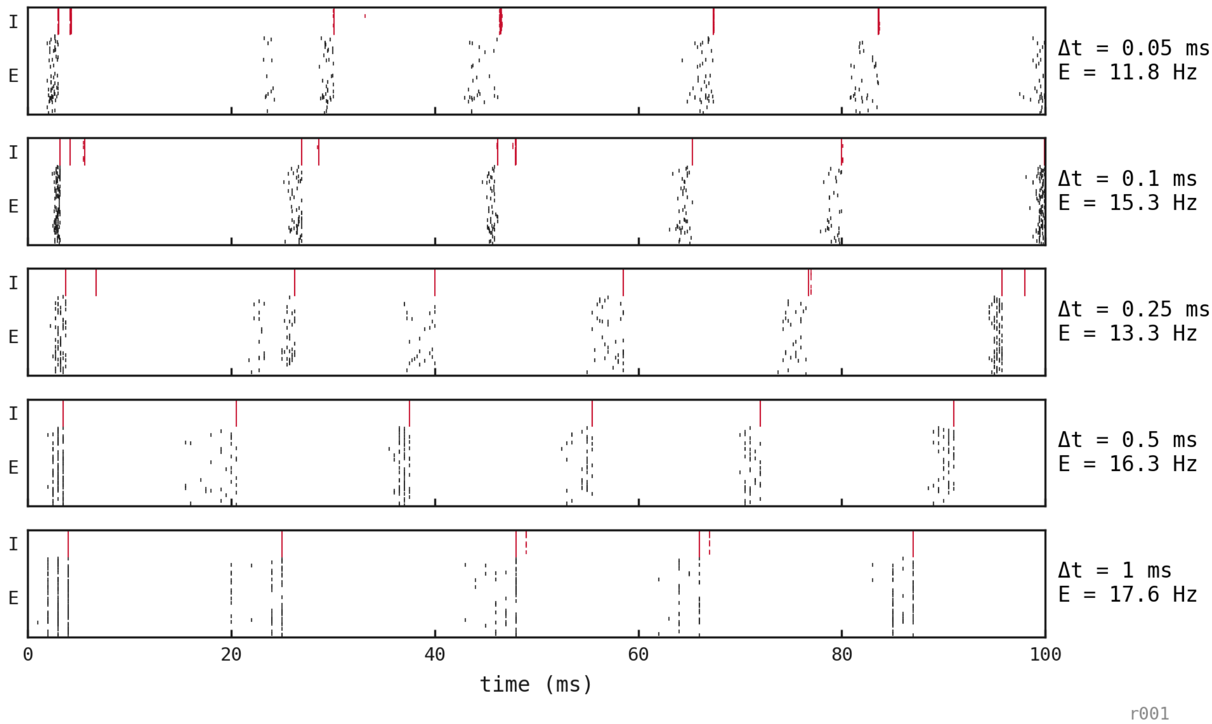
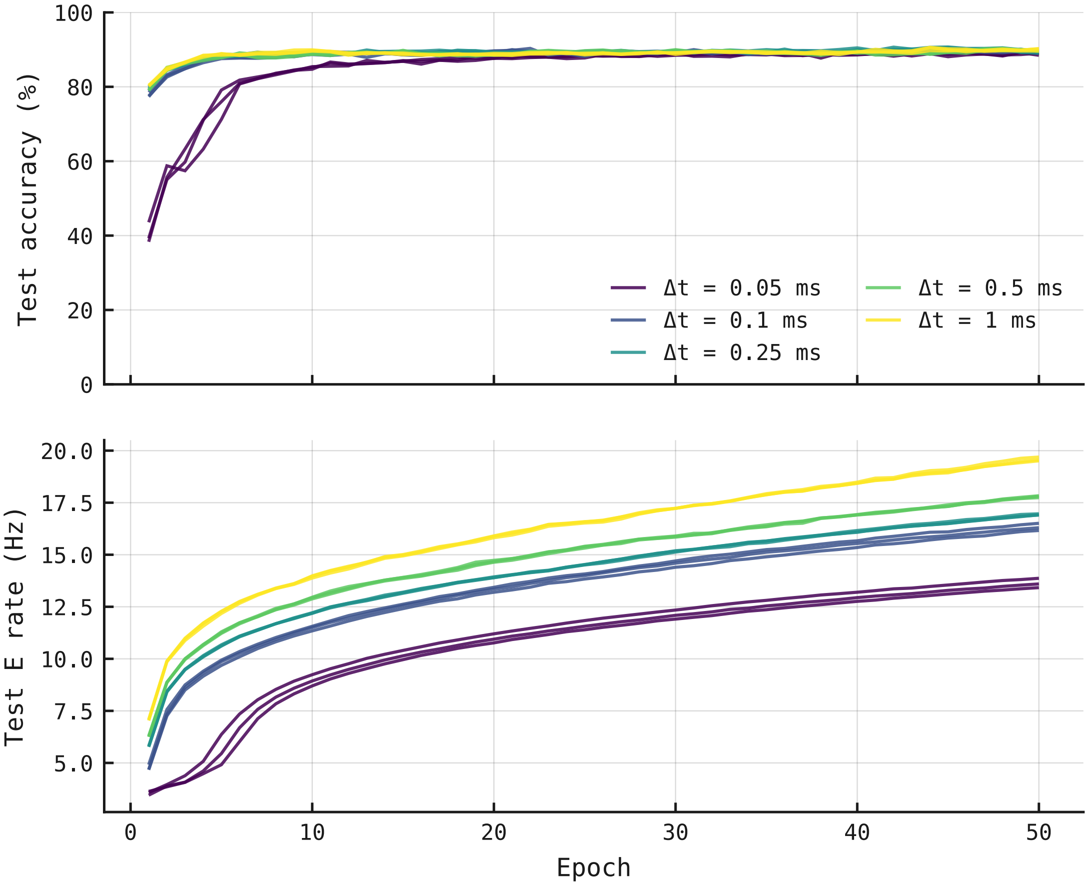


Figure 2: Single-trial rasters at each Δt, x-axis in *physical* ms (not steps). All five panels show E (black) and I (red) bursts locked to the same gamma cadence. The cycle physics is Δt-invariant.



r001

Figure 3: Top: test accuracy converges by epoch $\approx$ 10–15 across all Δt. Bottom: test E rate has *not* converged at epoch 50; every curve is still rising, ordered by Δt. The Δt-dependent rate is a snapshot at epoch 50, not a fixed-point ceiling.